

Francesco Mignone



 Turin

 fmignone98@gmail.com

 +39 331 905 9951

 GitHub

 LinkedIn

 ilnerdchuck.github.io

Programming Languages

C
Assembly x86/RISC-V/ARM
SQL

HDL Languages & Tools

SV/VHDL/UVM
Modelsim/Questasim
Design Compiler
Innovus
Quartus Prime
Z01X/VC-Z01X

Scripting

TCL
Bash/Shell

Embedded Systems & Tools

FreeRTOS
Arduino
Raspberry Pi
STM32

Softwares

Linux
Docker
KiCAD
GDB
Git

Technical Skills

Soldering
Logic Analyzers
PCB design
3D Printing & Design
LaTeX
Markdown

Spoken Languages

Italian - Native
English - C1 IELTS

For more information's on the projects or me visit my website!

EDUCATION

2024
March 2027

Polytechnic University of Turin: Masters Degree in Embedded Systems

 Turin

Coursework: Microelectronics Systems, Synthesis and Optimization of Digital Systems, Electronics for Embedded Systems, Specification and Simulation of Digital Systems, Software Engineering, Cybersecurity For Embedded Systems, Operating Systems for Embedded Systems, Computer Architectures (RISC, ARM), Integrated System Architecture, Testing and Fault Tolerance, Electronic Systems Engineering

Currently Remaining: Energy Management for IoT, Embedded Architectures for AI and ML

2017
2024

University of Pisa: Bachelor Degree in Computer Engineering

 Pisa

Relevant Coursework: Data Structures and Algorithms, Digital Logic Design, Computer Architectures (x86) and Kernel programming, Operating Systems, Electronic Circuits Analysis, Digital Electronics, Analog and Digital Communications, Computer Networks, Operations research, Automation Engineering

Thesis: Extension for VSCode to debug a multi-programmed Kernel

JOBS AND INTERNSHIPS

2026
Today

ARM - Internship and Master's Thesis: Functional Safety and Cybersecurity

 Sophia Antipolis

VC-Z01X, SystemVerilog, Tcl, Scripting

Dependability analysis of ARM architecture via fault injection.

PROJECTS

2025

DLX Microprocessor

VHDL, Questasim, Design Compiler, Innovus

RTL Design, Simulation, Synthesis and Physical Design of a DLX Microprocessor

- Pipeline implementation: Control Unit and Datapath
- Subset of DLX ISA: Load, Store, Arithmetic, Logic, Branch and Jump instructions
- Full ASIC design flow: RTL Design, Simulation, Synthesis and Physical Design
- Developed custom scripts to automate the Simulation, Synthesis and Physical Design process

2025

NXP S32K on QEMU

C, QEMU, FreeRTOS

Implementation of the NXP S32K3X8EVB board family in wemu with a set of emulated features:

- UART, SPI peripheral, TPM module for secure boot and secure key storage
- FreeRTOS porting for test and drivers for the peripherals

2025

Verification Workshop

System Verilog, Questasim

Hands-on sessions of SystemVerilog and UVM for hardware design and verification to test basic designs

2025

Low-Power Contest

TCL, Prime Time, Design Compiler

TCL script for post Synthesis analysis to optimize the leakage power of a design with a Multi- V_{th} approach

2024 - Today

GBA-SPI

KiCAD, RaspberryPi, Arduino

- Schematic and PCB design
- LiPo battery protection and charger circuit
- External custom cartridge modules via GPIO
- ATmega power handling and I²C communication with a RaspberryPi
- Custom RaspberryPi I²C Driver for the ATmega